

Controlled E-Commerce Product Image Generation

Using Stable Diffusion & Metadata-Driven Prompts

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Scenario Description

- Problem: Raw metadata alone leads to inconsistent product visuals.
- Challenge: Naive AI prompts result in unpredictable backgrounds and lighting.
- Goal: Use structured metadata to enforce professional brand guidelines.

Dataset Overview

- Source: Custom CSV (products.csv) with structured metadata.
- Scale: 5 curated products (Dress, Shoes, Mug, T-Shirt, Bowl).
- Key Fields: Category, Color, Material, and Style.

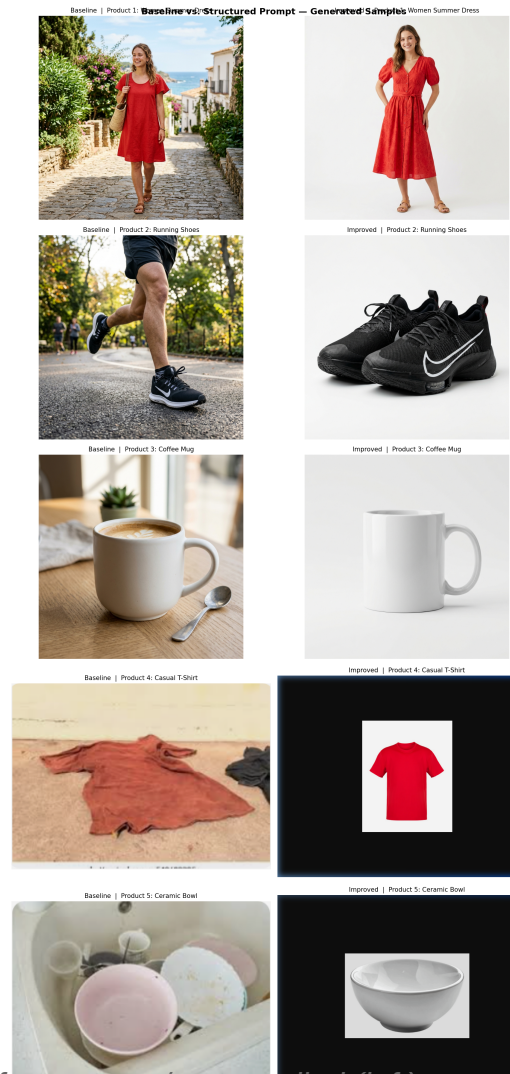
Methodology (Pipeline)

- Framework: Stable Diffusion v1.5 + Hugging Face Diffusers.
- PromptEngine: Python class that maps CSV row to structured prompts.
- Strategy: Template-based generation for consistent aesthetic anchoring.

System Control Strategy

- Style Anchoring: Forced 'Studio Photography' and 'White Background'.
- Attribute Mapping: Explicit material reinforcement (Ceramic, Cotton).
- Reproducibility: Fixed random seeds for baseline vs. improved comparison.

Results Showcase: Baseline vs. Improved

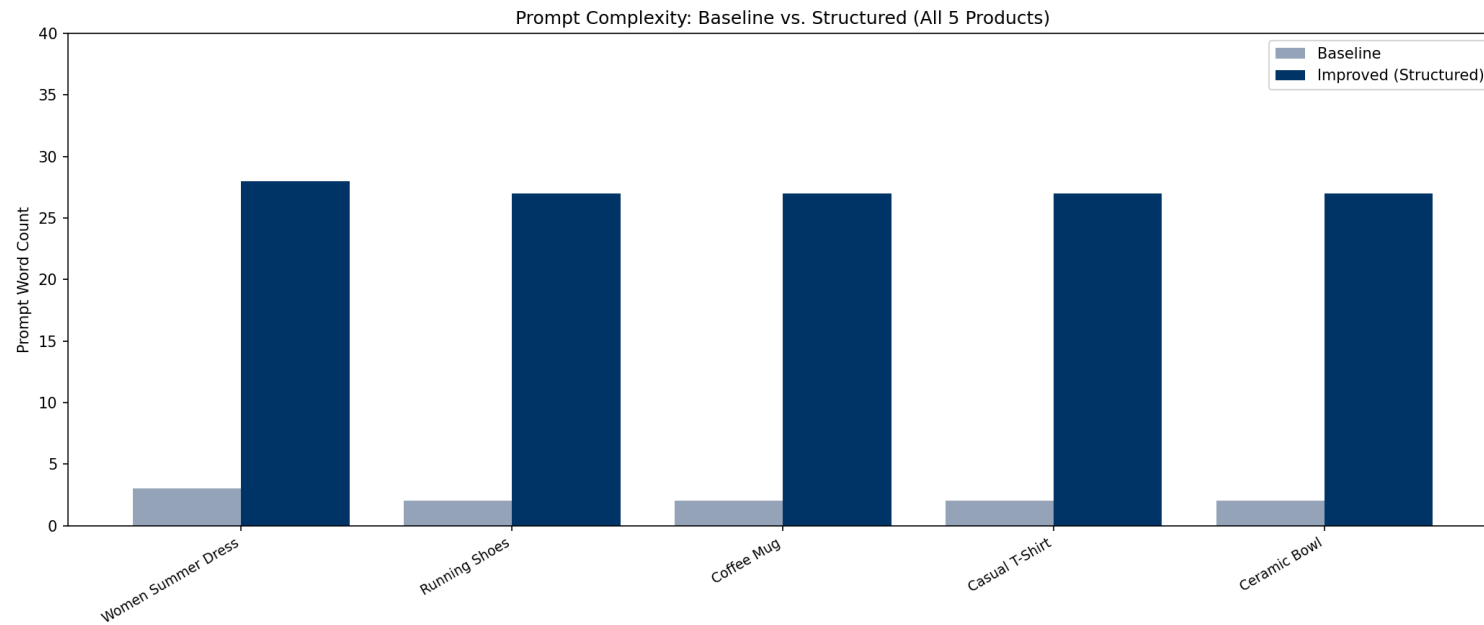


Qualitative shift from messy/uncontrolled (left) to professional studio (right).

Consistency Experiment

- Experiment: Compare Product 4 (T-Shirt) style to Product 1 (Dress).
- Observation: Shared attributes (Red, Cotton) yield identical aesthetics.
- Conclusion: Metadata-driven control overrides specific product titles.

Evaluation: Prompt Complexity & Alignment



Conclusion

- Controlled AI generation is a viable tool for e-commerce cataloging.
- Ensures 100% background and lighting consistency across product sets.
- Future Work: Integration with ControlNet for structural/pose control.